

Adaptive Testing for Data Science Recruitment

DSAN 6550: Adaptive Measurement with AI

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27 April 2026



GEORGETOWN UNIVERSITY



Overview of Adaptive Test



What it is?

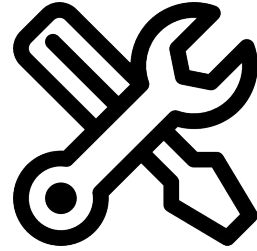
A 30 item adaptive
test for recruiting a
junior data scientist
role

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How it works?

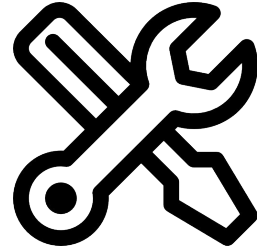
Each question targets the candidate's latent ability level

Overview of Adaptive Test



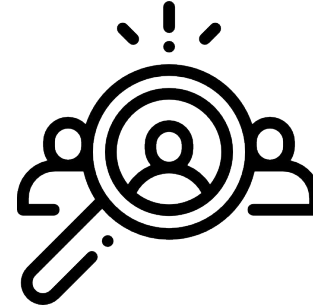
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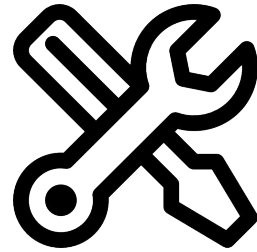
A precise ability score as well as performance across topics
Can compare different candidates

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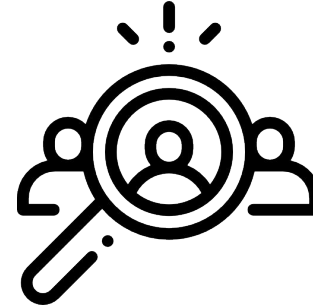
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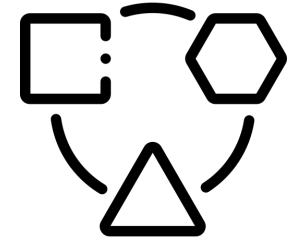
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How is it adaptive?

Questions across three difficulty levels. Each new question based on candidate's prior response.

Project Workflow (I): Item Development



Hi chat, I am a data science recruiter and I am recruiting for an entry level data scientist position. I want you to create an item bank of 30 questions for me which I can use in an adaptive test as part of my recruitment. I want to employ a 3PL model later for the adaptive testing, so they should differ on the basis of difficulty, discrimination and guessing. They should be multiple choice, five choices each.

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Item 1 — Statistics

Stem: Which of the following is a measure of central tendency?

- A) Standard deviation
- B) Range
- C) Median
- D) Variance
- E) Interquartile range

Correct Answer	a (discrim.)	b (difficulty)	c (guessing)	Topic
C	1.20	-2.10	0.22	Statistics

Rationale: Very easy definitional item; most candidates know median is a central tendency/measure.

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Item 18 — Machine Learning

Stem: Which statement BEST describes the role of the learning rate in gradient descent?

- A) It determines how many features the model uses
- B) It controls the step size taken in the direction of the negative gradient at each update
- C) It sets the number of training epochs
- D) It determines the regularization strength
- E) It defines the activation function

Correct Answer	a (discrim.)	b (difficulty)	c (guessing)	Topic
B	1.95	0.35	0.19	Machine Learning

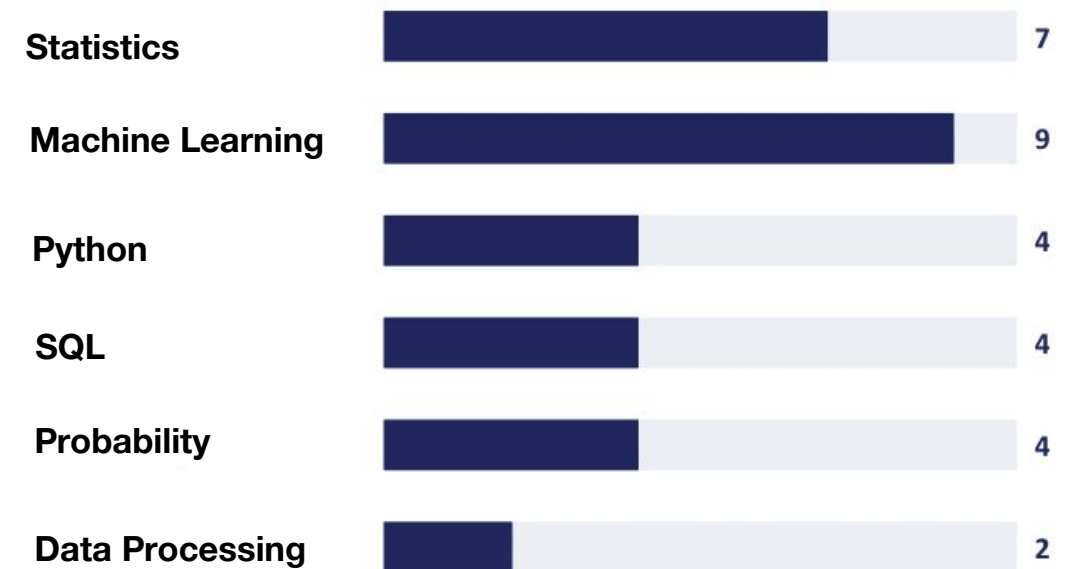
Rationale: Conceptual understanding of optimization.

Project Workflow (I): Item Development

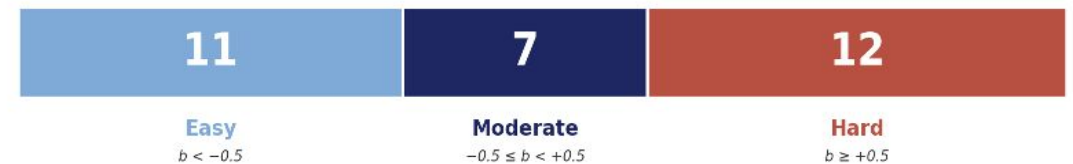


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Topic Distribution



Difficulty Distribution



Project Workflow (II): Response Generation



Ok now I want you to generate response data for each of these questions for 520 candidates. Give that to me in terms of a CSV file. Before you proceed, should I get the data in correct/incorrect or in terms of the response, i.e. option a, b, c, etc.

Project Workflow (II): Response Generation



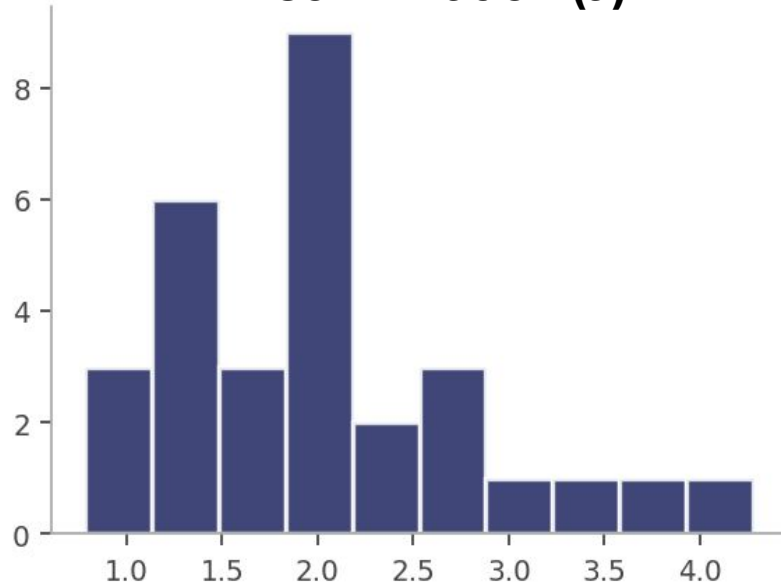
Ok now I want you to generate response data for each of these questions for 520 candidates. Give that to me in terms of a CSV file. Before you proceed, should I get the data in correct/incorrect or in terms of the response, i.e. option a, b, c, etc.

candidate_id	theta_true	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
C1000	-0.79	1	1	1	1	1	1	1	0	1	1
C1001	0.24	0	1	1	1	1	1	1	1	1	1
C1002	-1.90	1	1	0	1	1	1	0	0	0	0
C1003	1.40	1	1	1	1	1	1	1	1	1	1
C1004	0.64	1	1	1	1	1	1	1	1	1	1
C1005	-0.29	1	1	1	1	1	1	1	1	1	0
C1006	-0.31	1	1	1	1	0	1	1	1	1	1
C1007	0.30	1	0	1	1	1	1	1	1	1	1

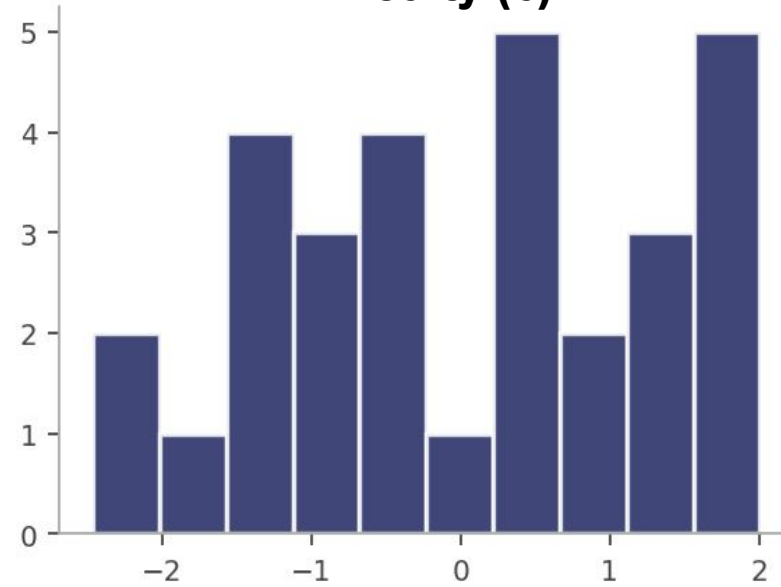
Project Workflow (III): Item Calibration

$$\text{3PL Model: } P(\theta) = \mathbf{c} + (1 - \mathbf{c}) / (1 + \exp(-1.7 \cdot \mathbf{a} \cdot (\theta - \mathbf{b})))$$

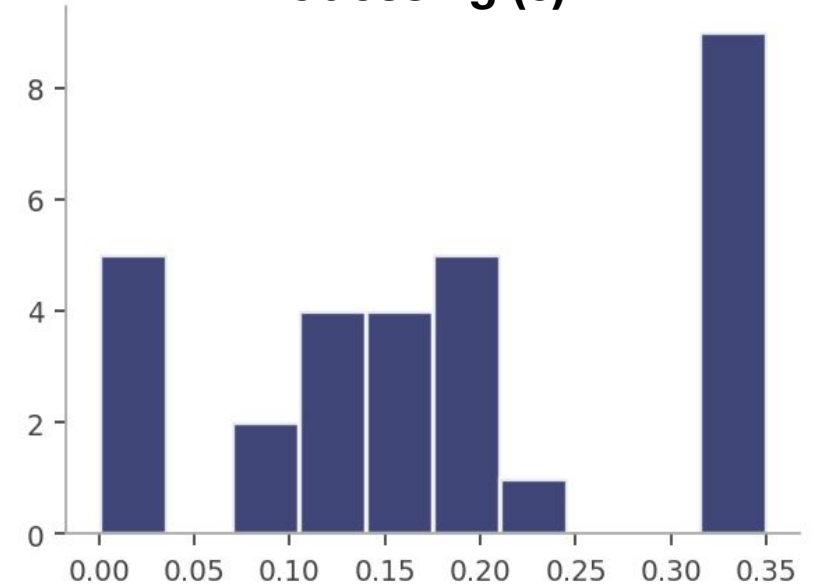
Discrimination (a)



Difficulty (b)



Guessing (c)



Project Workflow (III): Item Calibration

Reliability

0.84

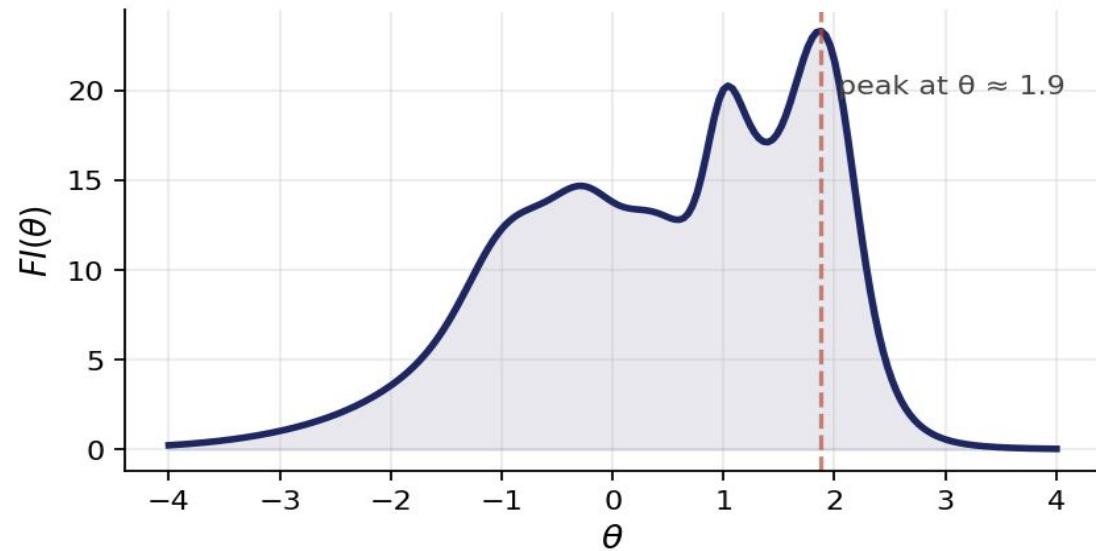
Cronbach alpha

Validity

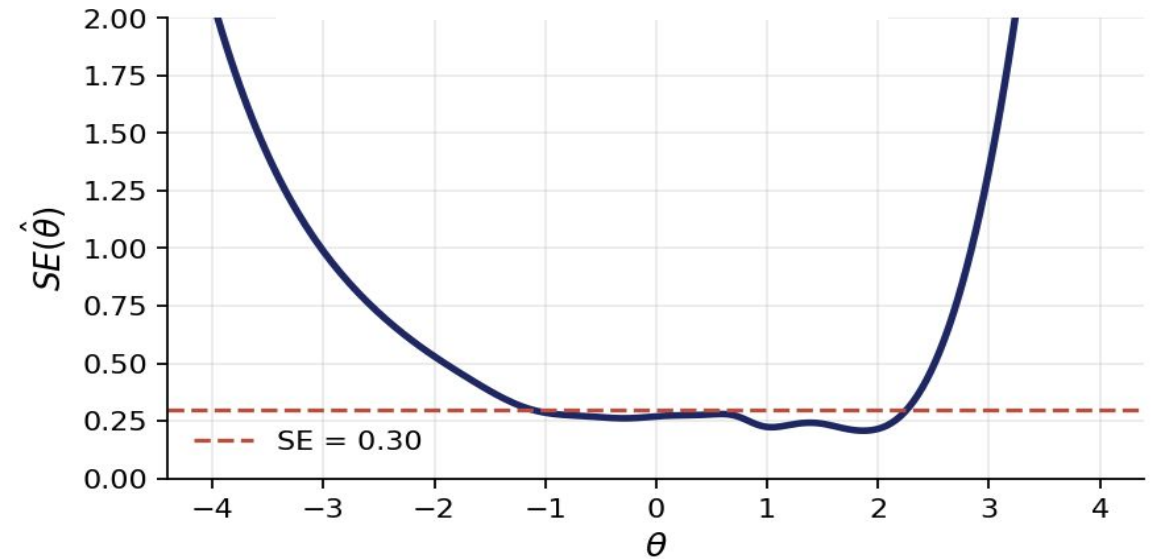
0.91

Correlation of Theta MLE

Test Information Curve



Standard Error Curve



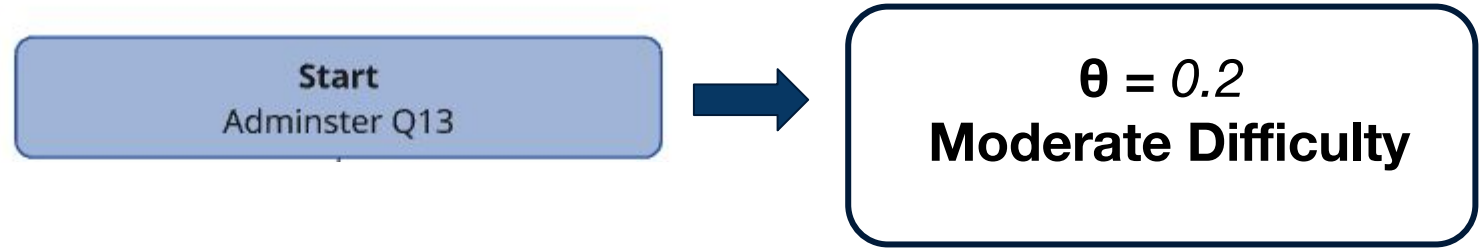
Project Workflow (IV): Adaptive Testing Algorithm

QUESTION 1 OF UP TO 20

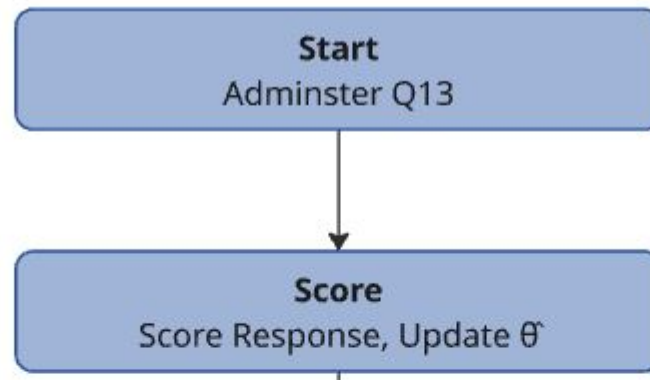
High variance leads to:

- ☐ A. Underfitting
- ☐ B. Overfitting
- ☐ C. Low error
- ☐ D. High bias
- ☐ E. No error

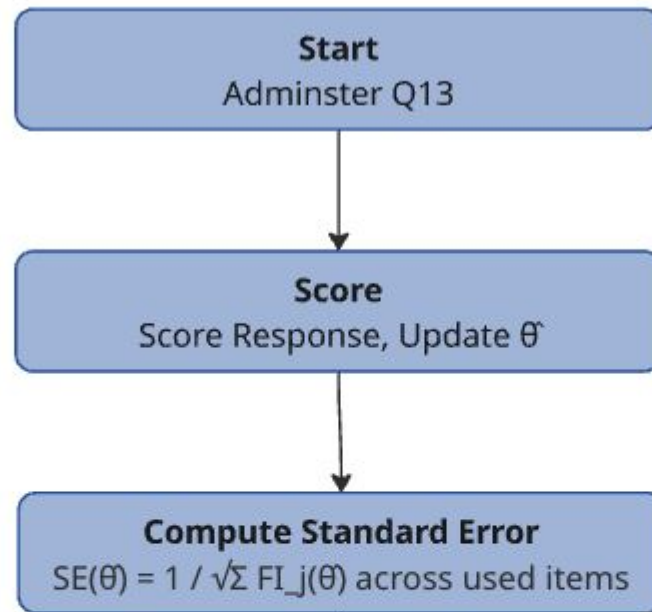
Submit Answer



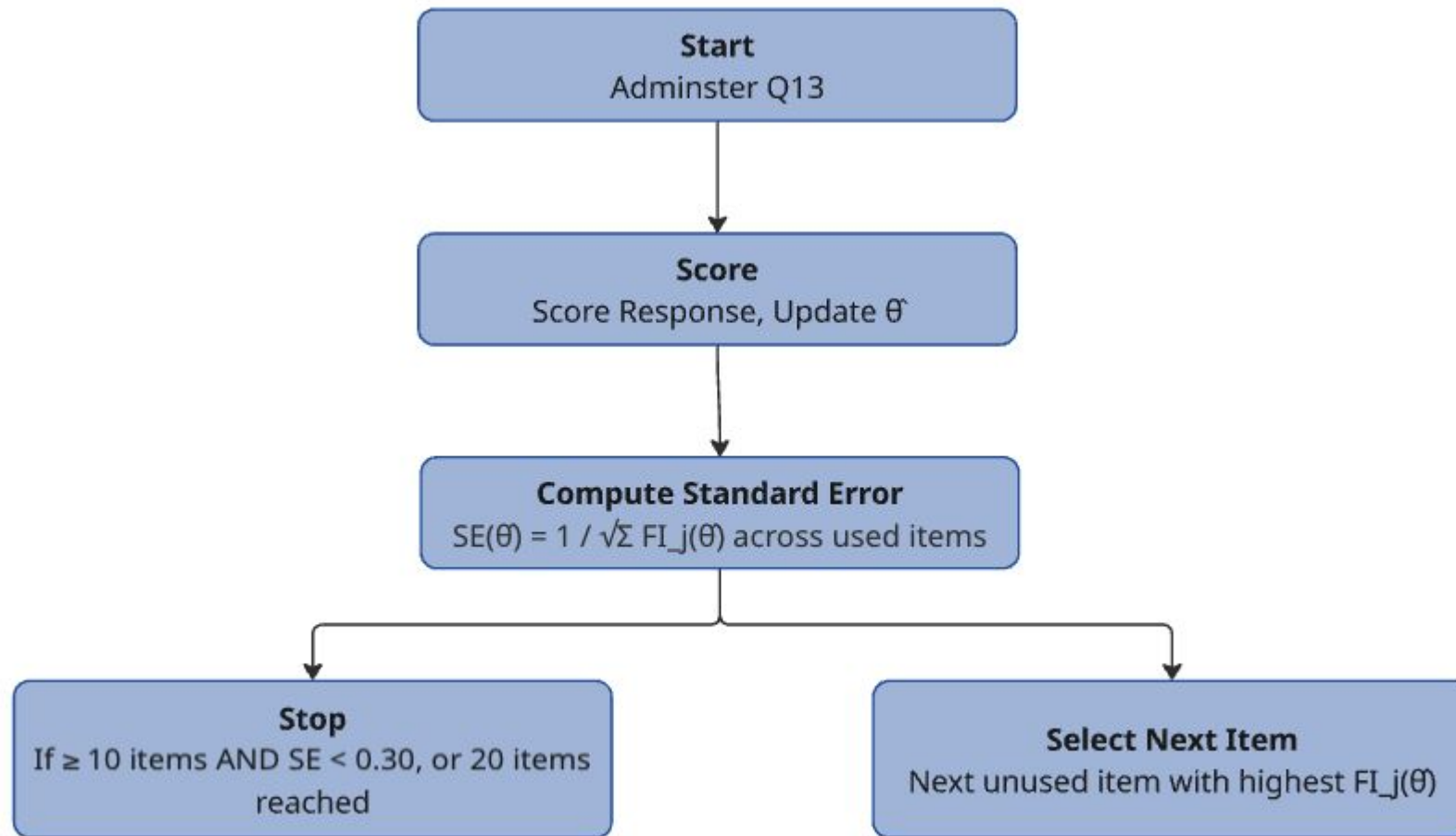
Project Workflow (IV): Adaptive Testing Algorithm



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Project Workflow (IV): Adaptive Testing Algorithm



Limitations (I)



Effectiveness

Highly effective in
distinguishing moderately
high ability candidates,
NOT low and very high
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Limitations (I)



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Generate more items
with extreme (very
high, very low) values
for difficulty
parameter

Limitations (II)



Early Test Instability

Initially (i.e. for first 2, 3 questions) θ estimate is very noisy. If candidate gets the first question wrong by accident, need few more questions for algorithm to correct itself

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Instead of just one question to set the initial θ benchmark, use at least 3 questions for a more robust estimate

Limitations (III)



No Control Over Topics

Algorithm maximizes only for information. Not sure if questions are cover all five DS topic areas

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Do stratified selection of items by topic instead (2 from SQL, Python, etc.)

Demo and Live Exam- Shiny App

<https://tinyurl.com/dsan-csm>

Data Science Recruitment Test (3PL Adaptive Testing) Demo Live Exam

CANDIDATE

Medium ability (theta_true = 0.01) ▲

Low ability (theta_true = -1.48)

Medium ability (theta_true = 0.01)

High ability (theta_true = +1.49)

Reset

STOPPING RULE

- Min items: 10
- Max items: 20
- $SE \leq 0.3$

ITEM HISTORY

(none)

CURRENT STATE

No items administered yet.

Click *Administer Next Item* to begin.

First item: **Q13**

ICC — CURRENT ITEM

No items yet

IIC — CURRENT ITEM

No items yet

STANDARD ERROR HISTORY

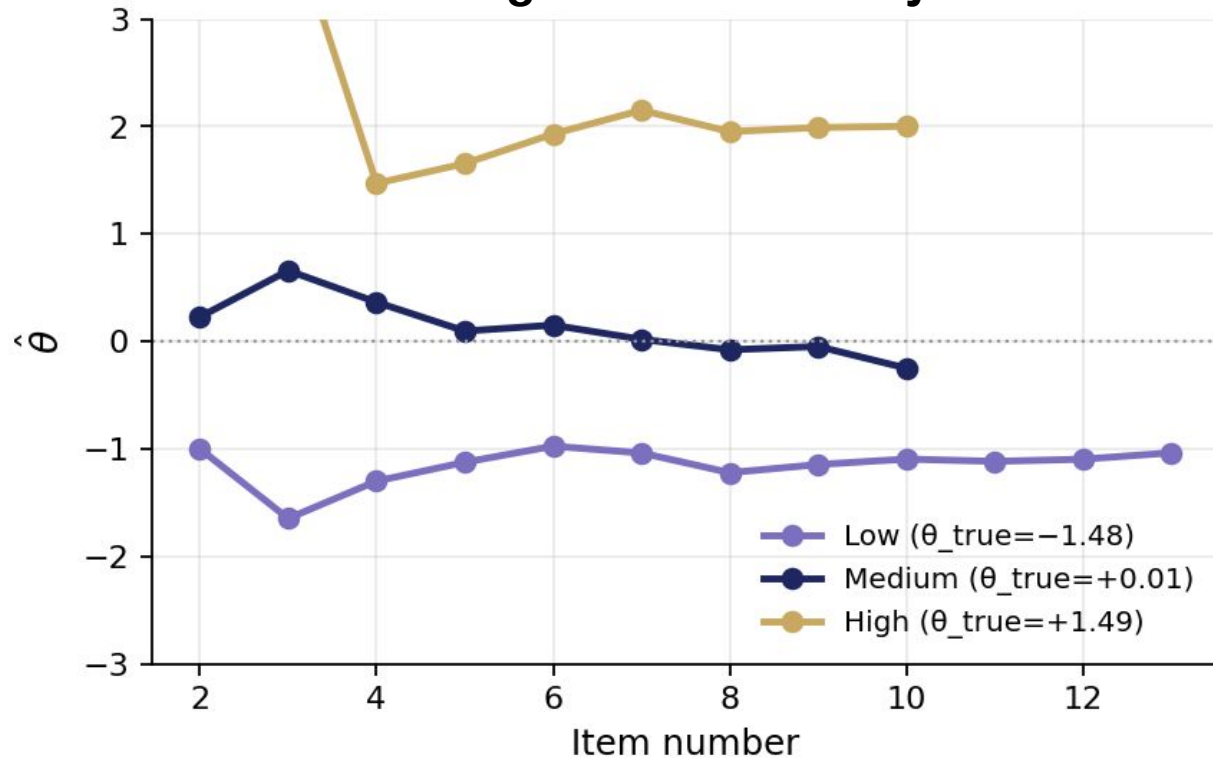
No items yet

LATENT TRAIT ESTIMATE HISTORY

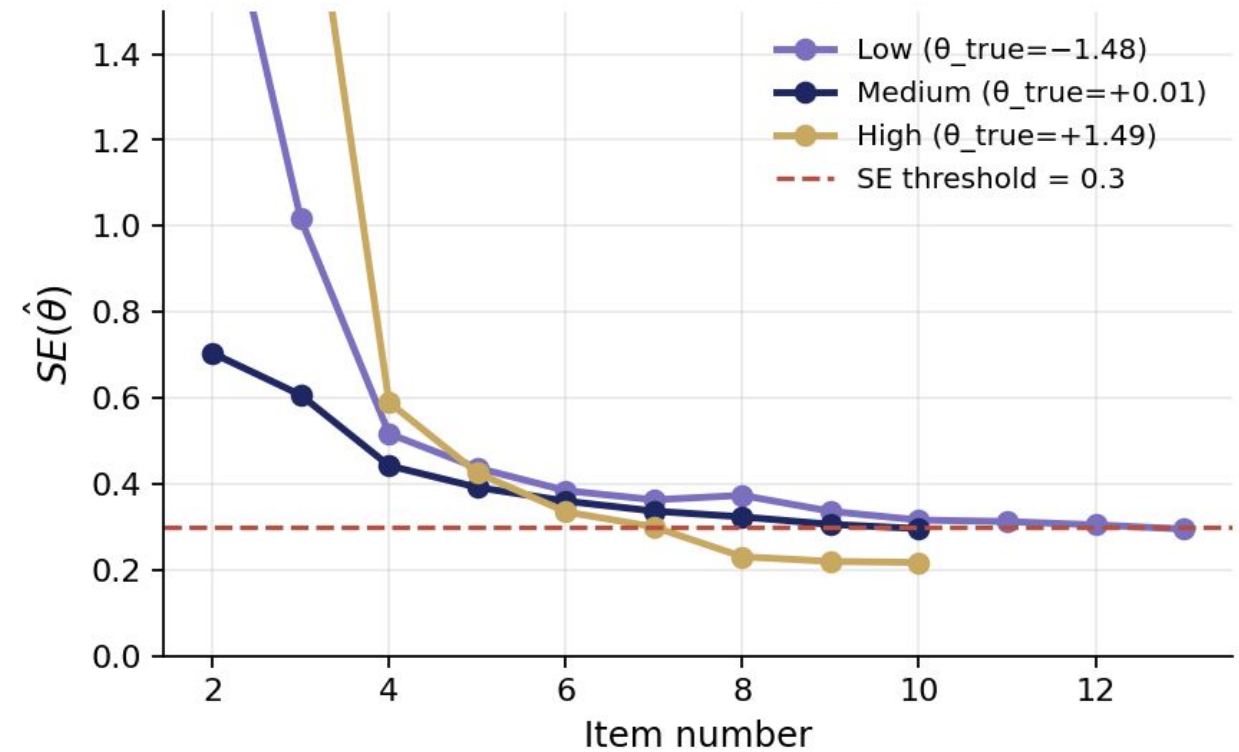
No items yet

Demo Results: Low, Medium, High Latent Ability

Change in Latent Ability θ



Change in Standard Error



Demo and Live Exam- Shiny App

Data Science Recruitment Test (3PL Adaptive Testing) Demo Live Exam

QUESTION 2 OF UP TO 20

PCA is used for:

- ☐ A. Classification
- ☐ B. Dimensionality reduction
- ☐ C. Clustering
- ☐ D. Regression
- ☐ E. Scaling

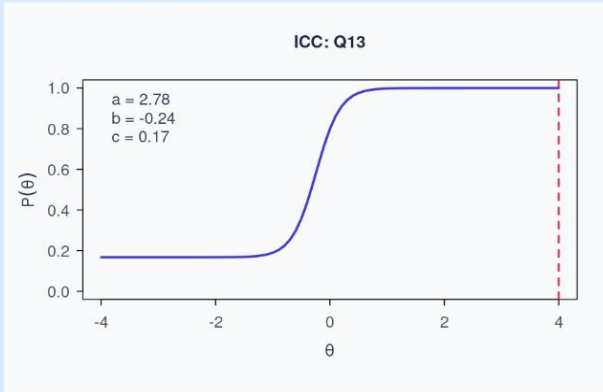
Submit Answer

ABILITY ESTIMATE θ

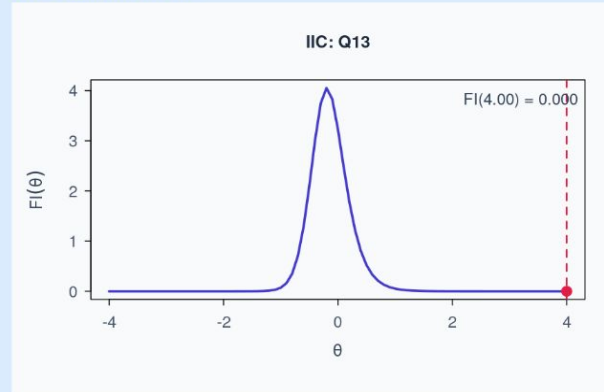
4.000 SE: 5242.169

Items answered: 1

ICC — CURRENT ITEM



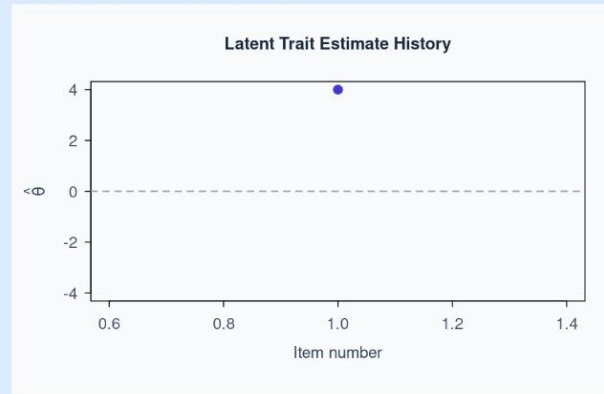
IIC — CURRENT ITEM



STANDARD ERROR HISTORY



LATENT TRAIT ESTIMATE HISTORY





Thank you!
Questions?